GradHunt AI: Multi-Agent Career Copilot Platform

Overview

GradHunt AI is an intelligent multi-agent career assistance platform designed to help job seekers discover opportunities, optimize application materials, prepare for interviews, and manage their job search workflow through autonomous AI agents.

Unlike traditional job boards that only aggregate listings, GradHunt AI acts as a personalized career copilot. The platform continuously analyzes user profiles, searches and ranks job opportunities, generates tailored application materials, tracks application progress, and provides interview coaching through coordinated AI agents operating on a shared memory and workflow orchestration framework.

The system combines large language models, retrieval-augmented generation (RAG), vector search, agent orchestration, and distributed task processing to deliver an end-to-end job search experience.

Objectives

The primary objectives of the platform are:

- Automate repetitive job search activities.
- Improve job matching accuracy through AI-powered analysis.
- Generate personalized application materials.
- Provide interview preparation and feedback.
- Maintain long-term career memory and user preferences.
- Demonstrate scalable multi-agent system design suitable for production environments.

System Architecture

The platform consists of five major layers:

Frontend Layer

Responsibilities:

- User authentication
- Resume upload
- Job dashboard
- Agent interaction interface
- Application tracking board
- Analytics and insights

Technologies:

- React
- TypeScript
- Tailwind CSS
- Redux Toolkit

API Gateway Layer

Responsibilities:

- Authentication
- Rate limiting
- Request routing
- Agent orchestration APIs

Technologies:

- FastAPI
- JWT Authentication
- API Gateway

Agent Orchestration Layer

Responsibilities:

- Workflow management
- Agent communication
- Tool invocation
- Shared memory access

Technologies:

- LangGraph
- OpenAI Function Calling

- Pydantic Models

Distributed Task Processing Layer

Responsibilities:

- Asynchronous execution
- Long-running workflows
- Retry management
- Scheduled jobs

Technologies:

- Redis
- Asynq
- Worker Pools

Data Layer

Responsibilities:

- User profiles
- Job postings
- Agent memory
- Application history
- Embedding storage

Technologies:

- PostgreSQL
- MongoDB
- pgvector
- Redis Cache

AI Agent Ecosystem

Job Discovery Agent

Purpose:

Identify relevant opportunities and rank them based on candidate fit.

Capabilities:

- Multi-source job crawling
- Duplicate detection
- Semantic matching
- Skill gap analysis
- Personalized ranking

Outputs:

- Match score
- Missing skills
- Salary estimation
- Application priority

Resume Optimization Agent

Purpose:

Improve resume quality for specific positions.

Capabilities:

- Resume parsing
- ATS compatibility analysis
- Keyword optimization
- Achievement enhancement
- Section-level recommendations

Outputs:

- ATS score
- Missing requirements
- Optimized bullet points
- Suggested revisions

Cover Letter Agent

Purpose:

Generate personalized application documents.

Capabilities:

- Company research
- Position analysis
- Resume integration
- Tone customization

Outputs:

- Tailored cover letters
- Networking messages
- Recruiter outreach drafts

Application Tracking Agent

Purpose:

Monitor application lifecycle and automate follow-up actions.

Capabilities:

- Application management
- Status prediction
- Follow-up reminders
- Timeline analytics

Stages:

- Saved
- Applied
- Online Assessment
- Recruiter Screen
- Technical Interview
- Final Round
- Offer
- Rejected

Outputs:

- Follow-up recommendations

- Success rate analytics
- Application health reports

Interview Coach Agent

Purpose:

Provide personalized interview preparation.

Capabilities:

- Question generation
- Mock interviews
- Behavioral coaching
- Technical assessment

Supported Topics:

- Data Structures
- Algorithms
- System Design
- Behavioral Interviews
- Company-Specific Questions

Outputs:

- Performance scores
- Improvement suggestions
- Knowledge gap reports

Agent Memory System

The platform implements long-term memory across all agents.

Memory Categories:

User Profile Memory

Stores:

- Skills

- Experience
- Education
- Career goals

Application Memory

Stores:

- Previous applications
- Interview results
- Recruiter interactions

Preference Memory

Stores:

- Preferred locations
- Compensation targets
- Company preferences
- Work style preferences

Learning Memory

Stores:

- Interview weaknesses
- Frequently missed concepts
- Historical feedback

Vector search enables agents to retrieve relevant memories during interactions.

Retrieval-Augmented Generation (RAG)

The platform utilizes RAG pipelines for contextual reasoning.

Knowledge Sources:

- Job descriptions
- Resume versions
- Interview notes
- Company information
- User history

Pipeline:

1. Query Generation
2. Embedding Search
3. Context Retrieval
4. LLM Reasoning
5. Structured Output Generation

Evaluation Framework

A dedicated evaluation system measures agent quality.

Metrics:

Resume Agent

- ATS Score Improvement
- Recommendation Acceptance Rate

Job Agent

- Precision@K
- Match Accuracy

Interview Agent

- Feedback Relevance Score
- User Satisfaction Score

Platform Metrics

- Latency
- Token Usage
- Cost per Request
- Success Rate

The evaluation framework continuously monitors agent performance and identifies failure patterns.

Monitoring and Observability

Production monitoring includes:

- OpenTelemetry tracing
- LangSmith workflow tracking
- Prometheus metrics
- Grafana dashboards

Tracked Metrics:

- Request latency
- Agent execution time
- Queue depth
- Error rates
- LLM costs

Cloud Infrastructure

Deployment Architecture:

AWS Services:

- EC2
- ECS Fargate
- API Gateway
- Lambda
- S3
- CloudWatch
- RDS PostgreSQL
- ElastiCache Redis

CI/CD:

- GitHub Actions
- Docker
- Terraform

Expected Scale

Target System Capacity:

- 10,000+ job records processed daily
- 5,000+ resumes analyzed monthly
- 1,000+ concurrent users
- Sub-second search latency
- 99.9% platform availability

Expected Outcomes

The platform is expected to:

- Reduce job search effort by 60%
- Improve application quality through AI-guided optimization
- Increase interview readiness through personalized coaching
- Provide a unified career management experience
- Demonstrate production-grade multi-agent system architecture suitable for enterprise-scale deployment
- **Agent Evaluation & Self-Improvement Pipeline** — agents automatically evaluate previous resume recommendations, interview feedback, and job matching outcomes to improve future decisions. This is currently one of the hottest topics in production AI systems and would make GradHunt stand out from typical "LLM + LangChain" projects.

Technology Stack

Frontend:
React, TypeScript, Tailwind CSS

Backend:
Python, FastAPI, Redis, PostgreSQL

AI/ML:
OpenAI API, LangGraph, RAG, Embeddings, pgvector

Infrastructure:
AWS, Docker, GitHub Actions, Terraform

Observability:
LangSmith, OpenTelemetry, Prometheus, Grafana